BME577: Drug Delivery **Spring 2025**
Ishaan Mehrotra
Dr. Fan Yuan

Project 4: Endocytosis of Drugs

1) *For $C_L = 0$, solve the steady state form of Equation 4 to determine the $[R_i]$ at steady state, and solve Equation 2 for the rate of new receptor synthesis V_s, assuming $[R] = 0.3 \times 10^5$ receptors per cell at steady state.*

1) Eq 4. $\frac{d[R_i]}{dt} = k_{eR}[R] + k_{eB}[B] - (k_{deg} f_R + k_{rec}(1-f_R))[R_i]$

$[R_i]_{ss}$ from Eq 4: At steady state, $\frac{d[R_i]}{dt} = 0$

$C_L = 0$, so $[B]_{ss} = 0$

$$0 = k_{eR}[R]_{ss} + k_{eB} \cdot 0 - (k_{deg} f_R + k_{rec}(1-f_R))[R_i]_{ss}$$

$$k_e[R]_{ss} = (k_{deg} f_R + k_{rec}(1-f_R))[R_i]_{ss}$$

$$\boxed{[R_i]_{ss} = \frac{k_{eR}[R]_{ss}}{k_{deg} f_R + k_{rec}(1-f_R)}}$$

Eq. 2. $\frac{d[R]}{dt} = -k_1 C_L [R] + k_{-1}[B] - k_e[R] + k_{rec}(1-f_R)[R_i] + V_s$

$$V_s = k_{eR}[R]_{ss} - k_{rec}(1-f_R)[R_i]_s$$

$[R]_{ss} = 0.3 \times 10^5$ receptors/cell

$k_{eR} = 0.03$ min^{-1}

$k_{deg} = 0.0022$ min^{-1}

$f_R = 0.2$

$k_{rec} = 0.08$ min^{-1}

$k_{deg} f_R + k_{rec}(1-f_R) = (0.0022)(0.2) + (0.08)(0.8) = 0.00044 + 0.064 = 0.06444$

$$[R_i]_{ss} = \frac{0.03 \times (0.03 \times 10^5)}{0.06444} = \frac{0.03 \times 3.0 \times 10^4}{0.06444} = \frac{900}{0.06444} \approx \boxed{1.4 \times 10^4 \text{ receptors/cell}}$$

$$V_s = k_{eR}[R]_{ss} - k_{rec}(1-f_R)[R_i]_{ss} = 0.03 \times (3.0 \times 10^4) - (0.08 \times 0.8) \times (1.4 \times 10^4)$$

$$\approx 900 - 896 = \boxed{4 \text{ receptors/cell/min}}$$

2) *Use the ordinary differential equation (ODE) solver in Python to validate that in the absence of the particle, the receptor levels on the cell surface and within the cells are constant if the initial conditions of [R], [R_i], and V_s are equal to the steady state values determined in Task 1.*

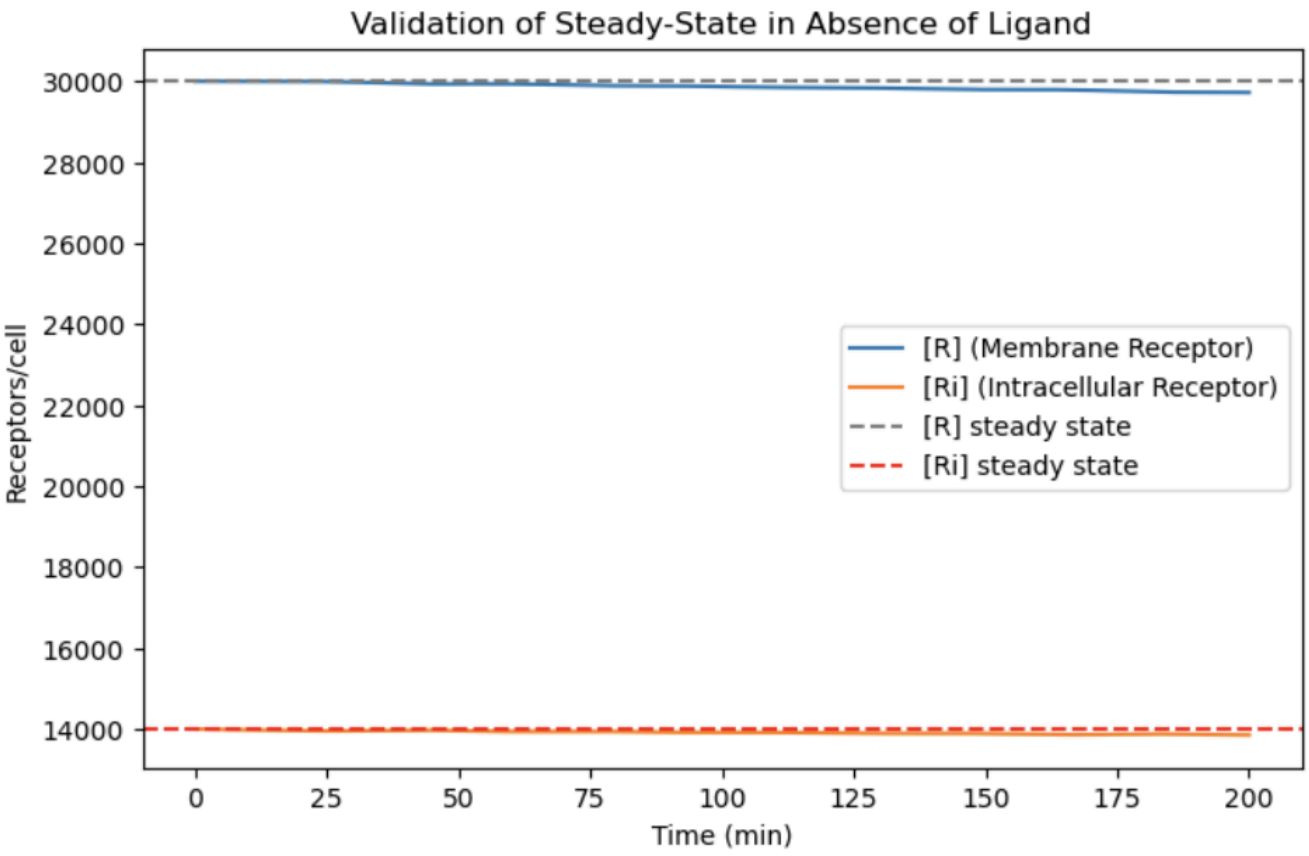


Figure 1. Time courses of membrane and intracellular receptor concentrations in the absence of ligand, showing that both [R] and [R_i] remain constant at their steady-state values over 200 minutes when initialized at steady state. This confirms the stability of the receptor system in the absence of extracellular particle.

This result confirms that the receptor system is at dynamic equilibrium when no ligand is present. The constant levels of membrane receptors [R] and intracellular receptors [R_i] over time indicate that receptor synthesis, internalization, recycling, and degradation are perfectly balanced. This steady-state behavior validates the model and parameter values in the absence of external stimulus.

3) *Cellular uptake of the particle will reduce C_L. The amount of reduction is proportional to the cell concentration or density (n_c). Determine the maximum n_c (n_{cmax}) at which the particle depletion, during a 30-min experiment, is negligible. In this homework, please try different values of n_c and find a value at which the change in C_L is between 0.9% and 1.0% of the initial C_L. Assume that (i) the initial values of [R], [R_i] and V_s are equal to the steady state values determined in Task 1, (ii) V_s is time-independent, and (iii) the initial value of C_L is 5 x 10^{-11} M.*

nc (cells / unit vol)	ΔCL / CL_0
2.488288e+08	0.9014%
2.498198e+08	0.9050%
2.508108e+08	0.9086%
2.518018e+08	0.9121%
2.527928e+08	0.9157%
2.537838e+08	0.9193%
2.547748e+08	0.9228%
2.557658e+08	0.9264%
2.567568e+08	0.9300%
2.577477e+08	0.9335%
2.587387e+08	0.9371%
2.597297e+08	0.9407%
2.607207e+08	0.9442%
2.617117e+08	0.9478%
2.627027e+08	0.9514%
2.636937e+08	0.9549%
2.646847e+08	0.9585%
2.656757e+08	0.9620%
2.666667e+08	0.9656%
2.676577e+08	0.9692%
2.686486e+08	0.9727%
2.696396e+08	0.9763%
2.706306e+08	0.9799%
2.716216e+08	0.9834%
2.726126e+08	0.9870%
2.736036e+08	0.9906%
2.745946e+08	0.9941%
2.755856e+08	0.9977%

Figure 2. Cell concentrations (n_c) that yield a 30‑min decrease in extracellular particle concentration (ΔCL) between 0.900 % and 1.000 %. The largest density satisfying this threshold—**2.76 x 10^8 cells per unit volume**—defines $n_{c,max}$, above which bulk depletion would exceed the 1% limit.

The grid search identifies every cell density that limits the 30‑minute decline in extracellular drug concentration, C_L, to between 0.9% and 1% of its initial value. Reading the table from lowest to highest densities, the final entry that still meets the criterion is $n_{c,max}$ = 2.8 x 10^8 cells per unit volume, at which the medium has lost 0.998% of its starting drug level. Any greater

density would deplete more than 1% of the drug, whereas lower densities remove less than 0.9%. Scientifically, this confirms that total particle uptake is proportional to the number of cells present: raising n_c raises cumulative internalization and thus lowers C_L. Establishing $n_{c,max}$ therefore defines a threshold below which the extracellular reservoir can be treated as effectively constant during a 30-minute experiment, ensuring that observed intracellular kinetics are not distorted by bulk depletion of the ligand.

4) *Use the math model and the parameter values listed in Table 1 to simulate receptor and particle concentrations (i.e., concentrations of R, B, R_i, L_i) as functions of time for initial value of C_L to be 5 x 10^{-11}, 5 x 10^{-10}, or 2 x 10^{-9}, respectively. For each initial concentration of the particle, solve the equations above using the ODE solver. The simulation period is 30 min, and $n_c = (n_c)_{max}$. In the simulation, verify that the maximal reduction in C_L in these cases is < 1%. Explain why.*

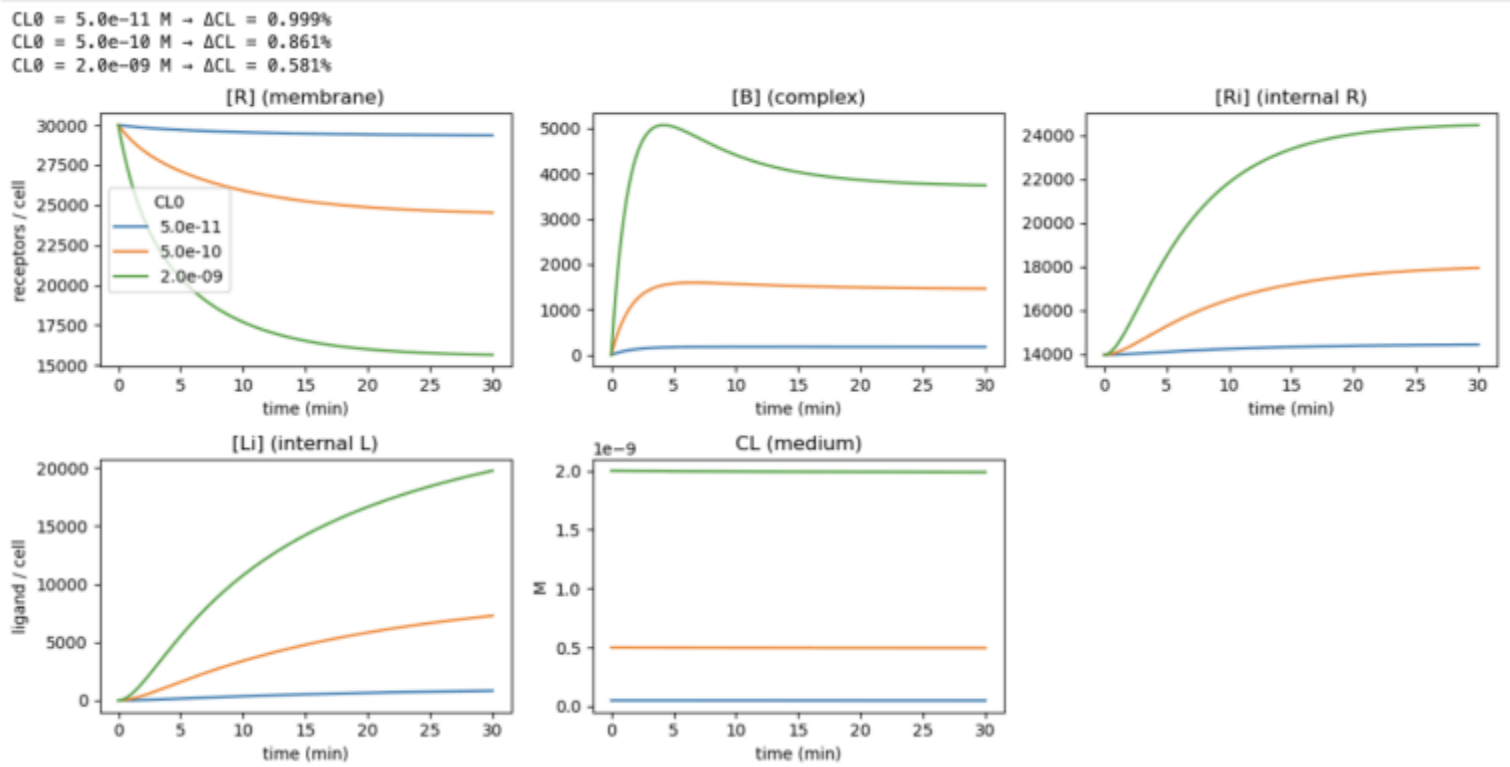


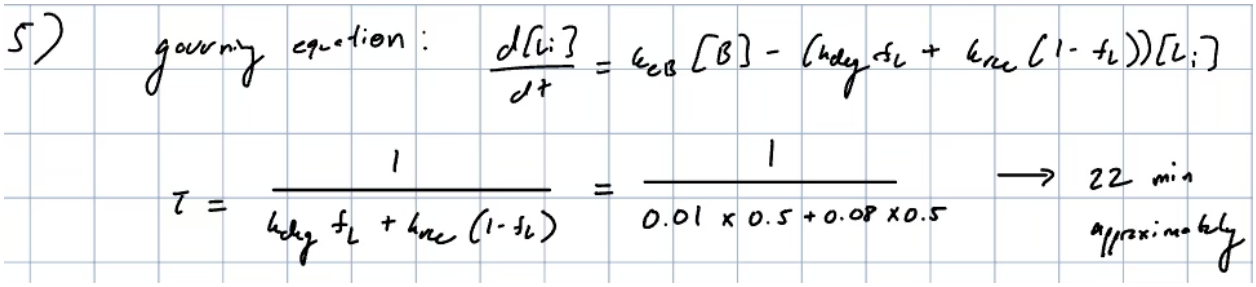
Figure 3. Time-course simulation of receptor-mediated uptake at n_c = 2.76 x 10^8 cells per unit volume. Panels show membrane receptors [R], surface complex [B], intracellular receptors [R_i], intracellular ligand [L_i], and extracellular ligand C_L over 30 min for three starting medium concentrations: 5 x 10^{-11} M (blue), 5 x 10^{-10} M (orange), and 2 x 10^{-9} M (green). Higher C_{L0} drives faster complex formation and larger intracellular accumulation, while the same cell density removes an almost constant absolute number of particles, so the fractional drop in C_L (values

printed above the plots) never exceeds 1%. Thus, the medium effectively acts as an infinite reservoir, allowing receptor and particle kinetics to be analysed without bulk-depletion.

The explanation here is simple and straightforward in verifying that the maximal reduction in C_L in these cases is less than 1%. The cell density was chosen so that, for the **lowest** starting concentration 5 x 10^{-11} M , cumulative uptake removes only 1% of the drug in 30 min. Because the absolute number of particles a cell can internalize is fixed by the kinetic parameters, increasing the initial reservoir (5 x 10^{-10} M and 2 x 10^{-9} M) makes the same uptake correspond to a smaller fraction of what is available outside. Hence, the percentage depletion falls well below 1% as $C_L(0)$ increases.

In other words, at $n_{c,max}$ the extracellular environment behaves like an infinite bath of sorts: it buffers the cells' uptake so thoroughly that bulk concentration changes are negligible, ensuring that intracellular dynamics we are observing reflect true receptor-mediated transport rather than artefacts of medium depletion.

5) *For the initial value of C_L at 2 x 10^{-9} M, approximately when will the intracellular concentration of the particle reach steady state?*



At the highest medium concentration tested, C_{L0} = 2 x 10^{-9} M, the intracellular drug pool $[L_i]$ rises rapidly for the first few minutes—driven by the burst of receptor–ligand complexes that forms on the surface—and then relaxes more slowly toward its asymptote as seen in the graphs in Figure 3 from Question 4. The governing equation above shows that once the surface complex [B] has basically stabilized, $[L_i]$ approaches its steady state value with a single-parameter time constant also calculated above.

For a first-order process, reaching 90% of the steady state takes about 2.3τ and 95% takes 3τ. Substituting the numerical τ found above gives approximately 50–65 min. This is consistent with the simulated trajectory (Figure 4): at 30 min $[L_i]$ is still climbing, having attained roughly three-quarters of its final level, whereas an additional half hour would bring it essentially to a plateau. Thus, the intracellular particle concentration is expected to reach steady state at roughly **one hour (60–65 min)** after exposure begins.